



Fig. 10: Krasnopol laser-guided projectile

photograph of Krasnopol projectile.

Krasnopol projectile follows a ballistic trajectory approaching the target once it is fired. Subsequent to its firing, a forward observer illuminates the target with a laser designator at a maximum range of 7km. The seeker in the front-end of the projectile locks on the target and the guidance and control system corrects its flight path in order to impact on the selected/illuminated point. Krasnopol projectile follows a top-attack pattern to achieve an optimised probability of kill against the armoured target.

M-712 Copperhead is a 155mm-calibre terminally laser-guided projectile with a minimum and maximum range of 3km and 16km, respectively. It has two operational modes: ballistic mode and glide mode. Ballistic mode is used when the cloud ceiling is high and visibility condition is good. Terminal guidance begins at 3km from the target. Glide mode is used when the cloud ceiling and/or visibility is too low to allow use of ballistic mode. The attack profile in the case of Copperhead projectile is laser-illuminated point attack. Fig. 11 shows Copperhead projectile in flight as it nears the target.

Copperhead projectile was successfully used during Operation Desert Storm in 1990-91 and Operation Iraqi Freedom in 2003. It is in use by various Armed forces, including Australian army, United States army, Egyptian army, Jordanian armed forces and Taiwanese army.

Laser-guided bombs

Of all the variants of laser-guided munitions, laser-guided bombs are the most widely exploited weapons if the number of user countries and if laser-guided bombs used in warfare in the past are any indication. Paveway fam-



Fig. 11: Attack profile of Copperhead projectile



Fig. 12: Paveway-II



Fig. 13: Paveway-IV

ily of laser-guided bombs has revolutionised tactical air-to-ground warfare by converting dumb bombs into smart precision-guided munitions. Paveway family of the laser-guided bomb is the preferred choice of Air Forces worldwide, as these have proven their accuracy and efficacy in almost all major conflicts in the past. The family has evolved over the years and has seen continuous capability enhancement with newer versions. It has seen four generations, namely, Paveway-I, -II, -II Plus, -III and -IV.

Paveway-I used a gimbaled seeker head, a computer control group (CCG) and a set of air foils. The seeker head operated in bang-bang mode, which meant that control surfaces were deflected either fully or not at all. This led to a sub-optimal flight trajectory. Bombs that could be fitted with Paveway-I LGB kit included M117, M118E1, MK-82, MK-83, MK-84, MK-20, CBU-74/B, CBU-75/B, CBU-79/B and CBU-80/B. More than 10,000 Paveway-I LGBs were used by the US Air Force in South East Asia with great success.

Paveway-II (Fig. 12) has a nose-mounted seeker head and fins for guidance. Manufactured by Defence contractors Raytheon and Lockheed Martin, it also uses the bang-bang guidance concept. That is, fins deflect fully or do not deflect at all.

Paveway-III is an improvement over Paveway-II and uses a more efficient proportional guidance technology.

Produced by Raytheon, it was introduced into service in 1983.

Paveway-IV (Fig. 13) is an advanced and highly accurate laser-guided weapon. It is the most recent member of the Paveway family. Manufactured by Raytheon Systems Ltd, the UK, Paveway-IV entered into service in 2008. It will replace Paveway II and enhanced Paveway-II weapon systems as well as the 453.6kg (1000-pound) unguided general-purpose bomb. Paveway-IV employs a combination of semi-active laser guidance and INS/GPS guidance to combine the flexibility and accuracy of laser guidance and all-weather capability of INS/GPS to give significantly improved battlefield performance.

Griffin laser-guided bomb kit is manufactured by Israel Aerospace Industries and is designed to retrofit the existing MK-82, MK-83 and MK-84 dumb gravity bombs. The kit employs a laser-seeker head and a set of steerable tail planes for guidance. The CEP is estimated to be 5m. It is in use by Israeli Defence Forces, Indian Air Force and Colombian Air Force.

Sudarshan laser-guided bomb kit developed by Aeronautical Development Establishment of DRDO and manufactured by Bharat Electronics is another LGB kit. It was introduced in Indian Air Force in 2013. The CEP is estimated to be 10m. In future, Sudarshan LGB kit will incorporate a GPS sensor to improve its performance.